EX-02-Cross-Platform-Prompting-Evaluating-Diverse-Techniques-in-AI-Powered-Text-Summarization

Prompt:

The Basics of Blockchain Technology

Blockchain technology is a decentralized digital ledger system used to record transactions securely and transparently across a network of computers. Unlike traditional databases that rely on a central authority, blockchain operates through a distributed network where each participant maintains a copy of the ledger. This structure reduces the need for intermediaries and increases trust among users. The main idea behind blockchain is to create a system where data can be recorded in a secure, tamper-resistant, and verifiable way.

A blockchain is composed of a sequence of blocks that store transaction data. Each block contains three main components: the transaction data, a timestamp, and a cryptographic hash of the previous block. The hash acts like a digital fingerprint, uniquely identifying each block. Because every block contains the hash of the previous block, the blocks are linked together in chronological order, forming a chain. If someone attempts to alter the data in a block, the hash changes, breaking the chain and making the tampering easily detectable.

One of the key features of blockchain technology is decentralization. In a decentralized network, control is distributed among many nodes (computers) rather than a single central authority. Each node participates in validating and storing transactions. This structure improves system reliability because the network can continue functioning even if some nodes fail or are attacked. It also increases transparency since all participants can view and verify the transactions recorded on the ledger.

To maintain accuracy and trust in the network, blockchain uses consensus mechanisms. These are protocols that allow network participants to agree on the validity of transactions before they are added to the blockchain. Two common consensus mechanisms are Proof of Work (PoW) and Proof of Stake (PoS). In Proof of Work, participants called miners solve complex mathematical puzzles to validate

transactions and create new blocks. In Proof of Stake, validators are chosen based on the amount of cryptocurrency they hold and are willing to “stake” as collateral. These mechanisms help prevent fraudulent activities such as double spending.

Blockchain technology is widely known for its role in supporting cryptocurrencies like Bitcoin and Ethereum. In cryptocurrency systems, blockchain enables secure peer-to-peer transactions without the need for banks or financial intermediaries. Every transaction is recorded on the blockchain and can be verified by anyone on the network, ensuring transparency and accountability.

Beyond cryptocurrencies, blockchain has many applications across different industries. In supply chain management, blockchain can track products from their origin to the final consumer, improving traceability and reducing fraud. In healthcare, it can securely store patient records and allow authorized professionals to access them when needed. In digital identity systems, blockchain can help individuals control and verify their personal information without relying on centralized authorities.

Despite its advantages, blockchain technology also faces challenges such as high energy consumption in some consensus mechanisms, scalability limitations, and regulatory concerns. However, ongoing research and technological advancements aim to address these issues and improve the efficiency of blockchain systems.

In conclusion, blockchain technology provides a secure, transparent, and decentralized method of recording and verifying transactions. Its unique structure and consensus mechanisms make it a powerful tool for building trust in digital systems, and its potential applications continue to expand across various sectors. summarize this paragraph.

OUTPUT:

ChatGPT:

Summary:

Blockchain technology is a **decentralized digital ledger** that securely records transactions across a distributed network of computers. Transactions are stored in blocks linked together using **cryptographic hashes**, making the data transparent and difficult to alter. The network relies on **consensus mechanisms** such as **Proof of Work** and **Proof of Stake** to validate transactions without a central authority. While blockchain is widely used for **cryptocurrencies like Bitcoin and Ethereum**, it also has applications in **supply chain management, healthcare, and digital identity systems**. Although it offers benefits like **security, transparency, and trust**, challenges such as **energy consumption, scalability, and regulation** still exist.

Gemini:

Blockchain technology is a **decentralized digital ledger** that records transactions across a network of computers, eliminating the need for a central authority. Each "block" in the chain contains data, a timestamp, and a unique **cryptographic hash** of the previous block, making the system tamper-resistant and transparent.

Key points include:

- **Security & Consensus:** The network stays accurate through mechanisms like **Proof of Work** (mining) or **Proof of Stake** (staking), which validate transactions and prevent fraud.
- **Decentralization:** Data is stored across many nodes, ensuring the system remains functional and verifiable even if parts of it fail.
- **Versatile Applications:** While famous for **cryptocurrencies** like Bitcoin, it is also used to secure medical records, track supply chains, and manage digital identities.
- **Current Hurdles:** The technology still faces challenges regarding **high energy use**, scalability, and regulation.

Claude:

Blockchain is a decentralized digital ledger that records transactions across networked computers in tamper-proof blocks linked by cryptographic hashes, ensuring security and transparency without central authorities. Key features include decentralization for reliability, consensus mechanisms like Proof-of-Work (miners solve puzzles) or Proof-of-Stake (validators stake crypto) to validate transactions and prevent fraud. Famous for powering cryptocurrencies like Bitcoin and Ethereum, it also applies to supply chains (traceability), healthcare (secure records), and digital identities. Challenges include energy use and scalability, but advancements continue to expand its potential.

Copilot:

Here's a clear, condensed summary of the passage:

Blockchain technology is a decentralized, tamper-resistant digital ledger that records transactions across a network of computers. It stores data in linked blocks containing transaction details, timestamps, and cryptographic hashes, making alterations easily detectable. Decentralization distributes control among nodes, enhancing reliability and transparency. Accuracy is maintained through consensus mechanisms like **Proof of Work** and **Proof of Stake**, which prevent fraud. While best known for powering cryptocurrencies such as Bitcoin and Ethereum, blockchain also supports applications in **supply chains, healthcare, and digital identity systems**. Despite challenges like energy use, scalability, and regulation, ongoing innovation continues to expand its potential.

Prompting Techniques:

Prompt:

The Basics of Blockchain Technology

Blockchain technology is a decentralized digital ledger system used to record transactions securely and transparently across a network of computers. Unlike traditional databases that rely on a central authority, blockchain operates through a distributed network where each participant maintains a copy of the ledger. This structure reduces the need for intermediaries and increases trust among users. The main idea behind blockchain is to create a system where data can be recorded in a secure, tamper-resistant, and verifiable way.

A blockchain is composed of a sequence of blocks that store transaction data. Each block contains three main components: the transaction data, a timestamp, and a cryptographic hash of the previous block. The hash acts like a digital fingerprint, uniquely identifying each block. Because every block contains the hash of the previous block, the blocks are linked together in chronological order, forming a chain. If someone attempts to alter the data in a block, the hash changes, breaking the chain and making the tampering easily detectable.

One of the key features of blockchain technology is decentralization. In a decentralized network, control is distributed among many nodes (computers) rather than a single central authority. Each node participates in validating and storing transactions. This structure improves system reliability because the network can continue functioning even if some nodes fail or are attacked. It also increases transparency since all participants can view and verify the transactions recorded on the ledger.

To maintain accuracy and trust in the network, blockchain uses consensus mechanisms. These are protocols that allow network participants to agree on the validity of transactions before they are added to the blockchain. Two common consensus mechanisms are Proof of Work (PoW) and Proof of Stake (PoS). In Proof of Work, participants called miners solve complex mathematical puzzles to validate transactions and create new blocks. In Proof of Stake, validators are chosen based on the amount of cryptocurrency they hold and are willing to “stake” as collateral. These mechanisms help prevent fraudulent activities such as double spending.

Blockchain technology is widely known for its role in supporting cryptocurrencies like Bitcoin and Ethereum. In cryptocurrency systems, blockchain enables secure peer-

to-peer transactions without the need for banks or financial intermediaries. Every transaction is recorded on the blockchain and can be verified by anyone on the network, ensuring transparency and accountability.

Beyond cryptocurrencies, blockchain has many applications across different industries. In supply chain management, blockchain can track products from their origin to the final consumer, improving traceability and reducing fraud. In healthcare, it can securely store patient records and allow authorized professionals to access them when needed. In digital identity systems, blockchain can help individuals control and verify their personal information without relying on centralized authorities.

Despite its advantages, blockchain technology also faces challenges such as high energy consumption in some consensus mechanisms, scalability limitations, and regulatory concerns. However, ongoing research and technological advancements aim to address these issues and improve the efficiency of blockchain systems.

In conclusion, blockchain technology provides a secure, transparent, and decentralized method of recording and verifying transactions. Its unique structure and consensus mechanisms make it a powerful tool for building trust in digital systems, and its potential applications continue to expand across various sectors. summarize this paragraph in Zero-Shot ,Few-Shot ,Chain-of-Thought ,Role-Based Prompt Technique

ChatGPT:

1. Zero-Shot Prompting

Summary:

Blockchain is a decentralized digital ledger that records transactions across a network of computers without a central authority. Transactions are stored in blocks linked by cryptographic hashes, making the data secure and tamper-resistant. The network verifies transactions using consensus mechanisms like Proof of Work (PoW) and Proof of Stake (PoS). Blockchain is widely used in cryptocurrencies such as Bitcoin and Ethereum and has applications in supply chains, healthcare, and digital identity systems. Despite advantages like transparency and security, it faces challenges such as scalability, energy consumption, and regulatory issues.

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2. Few-Shot Prompting

Summary:

Blockchain technology is a distributed and decentralized system that securely records transactions across multiple computers. Data is stored in blocks connected through cryptographic hashes, forming a tamper-resistant chain. Transactions are verified through consensus methods such as PoW and PoS, ensuring trust without a central authority. While blockchain powers cryptocurrencies like Bitcoin and Ethereum, it is also used in sectors like healthcare, supply chain management, and digital identity. The technology improves transparency and security but still faces challenges related to scalability and energy usage.

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3. Chain-of-Thought Prompting

Step-by-Step Reasoning Summary:

1. **Definition: Blockchain is a decentralized ledger used to record and verify transactions across a distributed network.**
2. **Structure: Transactions are grouped into blocks containing data, timestamps, and cryptographic hashes linking them together.**
3. **Security: The linking of blocks ensures that any data modification is easily detected.**
4. **Validation: Consensus mechanisms such as Proof of Work and Proof of Stake verify transactions before adding them to the chain.**
5. **Applications: Blockchain powers cryptocurrencies like Bitcoin and Ethereum and is also used in industries like healthcare and supply chains.**
6. **Conclusion: While blockchain offers transparency, security, and decentralization, it still faces challenges like scalability, energy consumption, and regulatory concerns.**

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4. Role-Based Prompting

Role: Technology Analyst

As a technology analyst, the article explains that blockchain technology is a decentralized system for securely recording transactions across multiple computers. It uses cryptographic hashing and distributed consensus mechanisms to ensure data integrity and transparency without relying on a central authority. The technology is most recognized for enabling cryptocurrencies such as Bitcoin and Ethereum. Beyond digital currencies, blockchain supports applications in supply chain tracking, healthcare data management, and digital identity verification. Although it provides strong security and trust, ongoing improvements are needed to address issues like scalability, energy usage, and regulatory frameworks.

Gemini:

1. Zero-Shot

Directly summarizing the content without prior examples.
Summary: Blockchain is a decentralized digital ledger that records transactions across a network of computers using linked "blocks" secured by cryptographic hashes. By utilizing consensus mechanisms like Proof of Work and Proof of Stake, it ensures security and transparency without a central authority. While it famously powers cryptocurrencies, its applications extend to supply chains and healthcare, though it still faces hurdles like scalability and energy use.

2. Few-Shot

Simulating a pattern based on examples (e.g., Topic: One-sentence summary).
Input Example 1: [Article about AI] -> AI mimics human intelligence to solve complex tasks.
Input Example 2: [Article about Solar Power] -> Solar power converts sunlight into electricity for sustainable energy.
Summary: Blockchain technology provides a secure, decentralized way to verify and store data across a global network.

3. Chain-of-Thought

Breaking down the logic step-by-step before reaching a conclusion.

- **Step 1 (Core Identity):** It is a decentralized, distributed ledger system.
- **Step 2 (Mechanism):** Blocks are linked by hashes, making them tamper-resistant.
- **Step 3 (Validation):** Consensus protocols (PoW/PoS) verify data without intermediaries.

- **Step 4 (Utility & Challenges):** It is used in crypto and industry but faces energy and scaling issues.
  Summary: Consequently, blockchain is a secure framework that uses cryptographic linking and collective validation to create a transparent, multi-industry tool for trustless data management.

4. Role-Based

Writing from the perspective of a specific persona.
Role: Tech Consultant for Beginners
"Think of blockchain as a shared digital record book that nobody can cheat. Instead of one bank keeping the books, everyone in the network has a copy. It uses clever math (hashes) to lock information in a chain, making it nearly impossible to hack. While it’s the engine behind Bitcoin, it’s also revolutionizing how we track products and protect medical data, despite some current 'growing pains' regarding speed and power usage."

Claude:

Zero-Shot Prompt

Prompt: "Summarize the article 'The Basics of Blockchain Technology' in 100 words or less, covering definition, structure, features, and applications."

Summary: Blockchain is a decentralized ledger recording transactions in linked blocks via cryptographic hashes for tamper-resistance. Key features: decentralization across nodes, consensus like PoW (mining puzzles) or PoS (staking). Powers crypto (Bitcoin, Ethereum); applies to supply chains, healthcare, identities. Challenges: energy use, scalability.

Few-Shot Prompt

Prompt:
"Example 1:
Input: Brief on AI.
Output: AI mimics human intelligence via machines learning from data.

Example 2:
Input: What is cloud computing?
Output: Cloud computing delivers on-demand resources over internet, scalable without owning hardware.

Now summarize: [full text]."

Summary: Blockchain is a decentralized ledger of transaction blocks linked by hashes, ensuring security without central control. Nodes validate via PoW or PoS

consensus; enables crypto like Bitcoin, plus supply chain tracking, health records, identities. Faces energy and scale issues.

Chain-of-Thought Prompt

Prompt: "Summarize the article step-by-step: 1. Identify main topic. 2. List key components (structure, features). 3. Note mechanisms and apps. 4. Mention challenges. 5. Combine into 1 paragraph."

Summary:

1. Main topic: Basics of blockchain as decentralized ledger.
2. Structure: Blocks with data, timestamp, prior hash; chain links them. Features: Decentralized nodes for reliability/transparency.
3. Mechanisms: PoW (puzzles), PoS (staking); apps: Crypto (Bitcoin/Ethereum), supply chain, healthcare, IDs.
4. Challenges: Energy, scalability, regs.
   Combined: Blockchain chains secure transaction blocks across decentralized nodes using PoW/PoS consensus, powering crypto and industries like supply chains despite energy hurdles.

Role-Based Prompt

Prompt: "You are a tech educator. Explain 'The Basics of Blockchain Technology' article to a beginner in a 150-word summary, using simple analogies."

Summary: Imagine blockchain as a shared notebook where everyone writes transactions in pages (blocks) locked with unique codes (hashes) linking to the previous page—no single boss in charge. Computers (nodes) copy the notebook for backup and vote on additions via puzzles (PoW) or bets (PoS) to keep it honest. Famous for Bitcoin cash swaps without banks; also tracks food from farm to table, secures health files, and manages IDs. It's super secure and open, but guzzles energy and grows slowly—fixes are coming!

Copilot:

◈ Zero-Shot

A direct summary without examples or prior guidance.

Summary:
Blockchain is a decentralized digital ledger that records transactions securely and transparently across a network of computers. It links blocks of data using cryptographic hashes, making tampering detectable. Consensus mechanisms like Proof of Work and Proof of Stake ensure trust and prevent fraud. While best known

for cryptocurrencies, blockchain also supports applications in supply chains, healthcare, and digital identity. Challenges include energy use, scalability, and regulation, but ongoing research seeks solutions.

◈ Few-Shot

Summarization after being shown examples of how summaries should look (simulated here).

Example Summaries (few-shot style):

- *Example 1:* "AI is reshaping industries by automating tasks and improving efficiency."
- *Example 2:* "Climate change is driven by human activity and requires urgent global action."

Summary (following the examples):
Blockchain is a secure, decentralized ledger system that links blocks of data with cryptographic hashes. It uses consensus mechanisms to validate transactions, supports cryptocurrencies, and has broader applications in supply chains, healthcare, and identity management. Despite challenges like energy consumption and scalability, it remains a transformative technology.

◈ Chain-of-Thought

Step-by-step reasoning before producing the summary.

Reasoning:

1. Blockchain is defined as decentralized and secure.
2. Its structure: blocks + hashes + timestamps.
3. Consensus mechanisms ensure trust (PoW, PoS).
4. Applications: cryptocurrencies, supply chain, healthcare, identity.
5. Challenges: energy, scalability, regulation.
6. Conclusion: powerful but evolving technology.

Summary:
Blockchain is a decentralized, tamper-resistant ledger that uses cryptographic links and consensus mechanisms to ensure trust. It underpins cryptocurrencies and offers applications in industries like supply chain and healthcare, though it faces energy and scalability challenges.

◈ Role-Based Prompt Technique

Summarization from different perspectives or roles.

- **Teacher:**
  "Blockchain is like a digital notebook shared across many computers. Each page (block) is linked to the previous one with a special code (hash), making it hard to cheat. People agree on what gets written using rules called consensus mechanisms. It's used in Bitcoin but also in healthcare and supply chains."
- **Business Analyst:**
  "Blockchain provides decentralized trust by eliminating intermediaries. Its applications extend beyond cryptocurrency into supply chain transparency, secure health records, and digital identity. Key challenges include scalability and regulatory frameworks."
- **Technologist:**
  "Blockchain is a distributed ledger secured by cryptographic hashes and consensus protocols such as PoW and PoS. It ensures immutability and transparency, enabling peer-to-peer transactions and industry applications, though energy efficiency and scalability remain pressing issues."

Would you like me to also **compare these techniques side-by-side in a table** so you can quickly see how the style shifts?

Platform Comparison:

Platform	Accuracy	Coherence	Simplicity	Speed	User Experience
ChatGPT	High	Very High	Very High	Fast	Excellent
Gemini	High	High	High	Very Fast	Good
Claude	Very High	Very High	High	Moderate	Excellent

Platform	Accuracy	Coherence	Simplicity	Speed	User Experience
Copilot	Medium	Medium	Medium	Fast	Good

Prompting Technique Comparison:

Prompt Technique	Strengths	Weaknesses
Zero-Shot	Fast, simple	May miss details
Few-Shot	Consistent structure	Requires examples
Chain-of-Thought	High accuracy and reasoning	Slightly slower
Role-Based	Most student-friendly summaries	Depends on role clarity

Conclusion:

The study shows that **prompt design significantly impacts AI output quality**. For educational content summarization:

- **Best Prompting Technique:** Chain-of-Thought + Role-Based
- **Best Platform:** ChatGPT or Claude

Combining structured reasoning with a clear educational role produces summaries that are **accurate, coherent, and easy for students to understand**.